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Major Article

Identifying and elucidating the resistance of *Staphylococcus aureus* isolated from hospital environment to conventional disinfectantsSamreen Sarwar M.Phil^{a,*}, Sidrah Saleem PhD^a, Faheem Shahzad PhD^b, Shah Jahan PhD^b^a Department of Microbiology, University of Health Sciences, Lahore, Punjab, Pakistan^b Department of Immunology, University of Health Sciences, Lahore, Punjab, Pakistan

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A B S T R A C T

Background: *Staphylococcus aureus* is a nosocomial pathogen, detection and elucidation of its resistance mechanisms to conventional disinfectants may aid in limiting its spread on environmental surfaces in health care settings. In the current study, disinfectant susceptibility of *S. aureus* strains isolated from the hospital environment as well as possible associations between the presence of disinfectant-resistance genes and reduced susceptibility to disinfectants was investigated.

Methods: A total of 245 samples were collected from the hospital environmental surfaces. The minimum inhibitory (MIC) and bactericidal concentrations (MBC) of disinfectants against *S. aureus* isolates were determined using the micro-broth dilution method. The qac genes (*qacA*, *qacE*, and *qacΔE1*) were detected by PCR and confirmed by sanger sequencing.

Results: A total of 47 *S. aureus* strains were isolated, with more than 85% of them showing methicillin resistance. The *qacA*, *qacE*, and *qacΔE1* genes were found in 23.4%, 29.7%, and 4.2% isolates respectively. All the isolates with qac genes had higher MIC and MBC values to selected disinfectants.

Conclusions: Significant methicillin resistant *S. aureus* (MRSA) contamination in the hospital environment was detected. Furthermore, higher qac gene frequencies were found in MRSA isolates that also correlated with higher MIC/MBC values to different disinfectants. The study proposes that hospitals should develop policies to determine disinfectant MICs against the common environmental isolates to contain the spread of resistant strains.

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INTRODUCTION

Hospital-acquired infections (HAIs) have been progressively rising over the last few decades, owing in part to the emergence of drug and disinfectant resistant bacterial strains, which are thought to have resulted from the misuse of antibiotics and disinfectants, respectively.^{1,2} Contaminated environmental surfaces in hospitals are known to play a significant role in the transmission of HAIs.³ The elimination of bacteria from the hospital environment is especially important for *Staphylococcus aureus* because these bacteria are easily disseminated through contact with surfaces on which they have been proven to persist for lengthy periods (up to 7 months).³ The public

now uses hospital-acquired Methicillin-resistant *Staphylococcus aureus* (MRSA) infection as a measure of hospital hygiene.⁴ *S. aureus*, as a nosocomial pathogen, can cause serious infections such as pneumonia, primary bacteremia, and infective endocarditis, particularly in immunocompromised patients.⁵

Cleaning and disinfection of environmental surfaces are critical elements of infection control and prevention programs in hospitals.⁶ Disinfectants are widely used in hospitals and are found in a wide range of products. Despite their growing use, bacteria are still sensitive to disinfectants when administered correctly. The widespread and indiscriminate usage of disinfectants in a variety of contexts, however, has prompted worries regarding the development of disinfectant-resistant bacteria.⁷ If disinfection failures occur due to disinfectant resistance and are not recognized by the relevant stakeholders, this can lead to serious repercussions such as outbreaks.⁸ Therefore, whether current disinfection regimens can achieve appropriate disinfection and eliminate nosocomial pathogens including the particular *S. aureus* strains

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found in the hospital environment is a pressing issue that must be addressed. In *S. aureus*, several plasmid-borne genes encoding efflux pumps confer reduced susceptibility to routinely used disinfectants in *S. aureus*, which warrants attention and investigation.²

In the present study, *S. aureus* strains were isolated from the environment of long-term care facilities of a tertiary care hospital in Lahore, Pakistan. The findings of this study will guide hospitals in terms of selecting disinfectants, preventing ineffective disinfection, containing the spread of disinfectant-resistant strains, and reducing the occurrence of HAIs.

METHODS

Specimen collection, isolation and identification of bacteria

It was a cross-sectional study that was approved by the Advanced Studies and Research Board (letter no. UHS/Education/126-20/898) and Ethical Review Committee (letter no. UHS/REG-20/ER/110) of the university, and Institutional Review Board (IRB) of the tertiary care hospital (letter no. SZMC/IRB/External/0028/2020). The sample size was calculated using data from a study by Cheatham et al.⁹ A total of 245 samples were collected between October 2021 and January 2022, from the environmental surfaces of medical wards (n = 4), general surgery and surgical oncology wards (n = 4), urology wards (n = 5), and orthopaedic wards (n = 2) at a tertiary care hospital in Lahore, Pakistan. These wards were chosen based on the type and epidemiology of nosocomial infections, hospital occupancy, and duration of stay.¹⁰

The sampling sites were categorized as follows: (i) Frequently touched surfaces (FTSs) (n = 64); these surfaces included IV pole grab area (n = 14), bedside table handles (n = 21), room light switches (n = 11), tray tables (n = 11), and room door knobs (n = 7). (ii) Medical equipment (ME) (n = 59); these included blood pressure apparatus (n = 12), cardiac monitor (n = 10), nebulizer (n = 1), oxygen supply unit (n = 29), surgical aspirator (n = 3), syringe pump machine (n = 1), weight machine (n = 1), and X-ray illuminator (n = 2). (iii) Patient's immediate surroundings (PISs) (n = 60); these surfaces included bed rails (n = 30), bed linen (n = 20), and patient chair (n = 10). (iv) Ward sinks (n = 62).

The sampling sites were selected based on existing data indicating that these sites pose the greatest risk of nosocomial infections. Similarly, the wards' handwashing sinks were sampled because these surfaces are known to harbor bacterial biofilms due to the moist and conducive environment.^{11,12} To collect samples, 30 cm² levelled environmental surfaces were swabbed for 1 minute with sterile swabs moistened with sterile 0.85% NaCl solution. Swabs were moved across the sampling area in a systematic manner that is, back and forth in 1 direction, then 90° degrees to the first, then diagonally. For unlevelled and protruding surfaces (eg, tubing and bed rails), as large an area as possible was swabbed.¹³

Upon completing the sampling, the swabs were placed in amies transport medium (Thermo Fisher Scientific, TS0001A) and were maintained at 2–8°C temperature during shipment to the lab. All specimens were inoculated on blood agar for 24 hours at 35°C. The *S. aureus* strains were identified using gram stain followed by catalase, slide coagulase, and DNase test.¹⁴

Antimicrobial susceptibility testing

Antibiotic Susceptibility testing was carried out by the Kirby-Bauer disk diffusion method according to the CLSI 2021 guidelines¹⁵ for linezolid (30 µg), penicillin (10 units), cefoxitin (30 µg), doxycycline (30 µg), trimethoprim-sulfamethoxazole (1.25/23.75 µg), ciprofloxacin (5 µg), clindamycin (2 µg), gentamicin (10 µg), erythromycin (15 µg), and azithromycin (15 µg). The susceptibility to vancomycin

and teicoplanin was tested using the standard micro-broth dilution method.¹⁶ For susceptibility testing, ATCC strains 25922 and 25923 were used as controls.

Determination of MICs and MBCs of disinfectants

The disinfectants used in this study were benzalkonium chloride (BAC) (Sigma, B6295), ~and 99% cetrimonium bromide (CTAB) (Sigma, H6269), and ≥99.5% chlorhexidine (CHX) (Sigma, 282227), because these are some of the most frequently used disinfectants in health care settings.¹⁷ For all *S. aureus* isolates, the MIC of each disinfectant was determined.¹⁸ In the absence of any standard breakpoints for disinfectants against *S. aureus*, concentrations modified from the data, reported by Vijayakumar and Sandle were used.² For MIC testing, the concentration ranges were 2–512 µg/ml for CHX, 1–256 µg/ml for BAC, and 4–1024 µg/ml for CTAB. The stock solution of BAC (1mg/ml) and CTAB (2mg/ml) was prepared in warmed distilled water. The stock solution of CHX (1mg/ml) was prepared in DMSO.

96-well microtiter plates were set up with doubling dilutions of the disinfectant with starting concentrations in Mueller Hinton (MH) broth. The bacterial suspension was prepared in 5ml sterile saline and diluted 1:100 times after being adjusted to a turbidity equivalent to that of a 0.5 McFarland standard. A fixed volume of the bacterial suspension was added to all wells with doubling dilutions of the disinfectants and positive control growth wells with no disinfectant. Two wells, 1 with distilled water and one with MH broth alone, were used as negative controls in each test. The final concentration of the inoculum was $\sim 1 \times 10^6$ CFU/ml in each well. After 24 hours of incubation at 35°C, the MIC values were recorded as the lowest concentrations of disinfectant that inhibited visible growth. The MBC of disinfectants was determined by inoculating 10 µl of the liquid from the wells with no growth onto tryptone soya agar (TSA) and incubating it for 24 hours at 37°C.¹⁸ MBC was defined as the lowest concentration of disinfectants resulting in the death of 99.9% or more of the initial inoculum.

Detection of *qac* genes in *S. aureus* isolates and sequencing

Genomic DNA from the *S. aureus* isolates was extracted using the boiling method.¹⁹ Multiskan Sky (Fisher Scientific Inc.) was used to quantify and assess the quality of extracted DNA samples. All *S. aureus* isolates were screened for the presence of the quaternary ammonium compound (*qac*) resistance genes *qacA*, *qacE*, and *qacΔE1* using PCR. Each PCR reaction consisted of 10 µL of DreamTaq Master Mix (2X) (Thermo Fisher Scientific, Waltham, MA USA), 7 µL sterile distilled water, 0.5 µL each of the forward and reverse primers (at 20 nM each), and 2 µL DNA to make up a total volume of 20 µL. The primer sequences for *qacA*, *qacE*, and *qacΔE1* genes were F5'-GCTGCATTATGACAATGTTTG-3' and R5'-AATCCACCTACTAAAGCAG-3', F5'-GCCCTACACAATTGGGAGA-3' and R5'-TTAGTGGG-CACTTGCTTTGG-3', and F5'-AATCCATCCCTGTCCGTTGTT-3' and R5'-CGCAGCGACTTCCACGATGGGGAT-3' respectively, as described by Hassan and Al-Khafaji.²⁰ Thermal cycling was performed on an iQ5 iCycler (Bio-Rad Laboratories, Inc., Hercules, California, USA) with annealing temperatures of 54.8°C, 57.2°C and 56°C for *qacA*, *qacE*, and *qacΔE1* genes respectively. The PCR-amplified products were detected by electrophoresis on a 2% agarose gel and viewed with a Bio-Rad Gel Doc XR+ Gel Documentation System (Bio-Rad Laboratories, Inc.). The amplicons obtained were purified and sequenced using the conventional Sanger dideoxy sequencing method at a commercial DNA sequencing service provider (Lab Genetix) for validation. DNA sequence data were analyzed by using NCBI-BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>).

Statistical analysis

Data were analyzed by SPSS v26. Frequency and percentages were calculated for drug resistance and *qac* genes. Chi-square was applied to determine the association between the carriage of disinfectant resistance genes and observable reduced antiseptic susceptibility. The MBC and MIC data were compared using the Mann-Whitney U-test and median test. *P*-value $\leq .05$ was considered statistically significant.

RESULTS

Distribution of *S. aureus* isolates in the hospital environment

A total of 47 (19.1%) *S. aureus* strains were isolated from 245 hospital environmental samples. The highest number of *S. aureus* isolates (25%) were obtained from orthopedic wards followed by urology wards (20.8%), medical wards (17.2%), and general surgery and surgical oncology wards (5.9%). Bedside table handles (38.1%) were found to be the most contaminated with *S. aureus* among the environmental surfaces, followed by bed rails (33.3%), oxygen supply unit (31%), and bed linen (25%). Table 1 shows the distribution of *S. aureus* at various sampling sites.

Antimicrobial susceptibility testing

Out of 47 isolates, 85% (*n* = 40) of the *S. aureus* isolates were resistant to cefoxitin and were identified as Methicillin-resistant *Staphylococcus aureus* (MRSA). Only 4 and 3 MSSA *S. aureus* isolates were found in the orthopedic and medical wards, respectively. Drug resistance manifested by *S. aureus* towards penicillin and erythromycin was 85.1%, followed by azithromycin (78.7%), ciprofloxacin (59.6%), clindamycin (34%), gentamicin (34%), trimethoprim/sulfamethoxazole (27.7%), doxycycline (14.9%), and linezolid (4.26%). Doxycycline, ciprofloxacin, trimethoprim/sulfamethoxazole, and teicoplanin susceptibility were intermediate in 10.6%, 4.3%, 2.1%, and 2.1% of *S. aureus* isolates, respectively. 79% of the isolates (*n* = 37) were multidrug-resistant (MDR), that is, they were resistant to 3 or more than 3 classes of antimicrobials. None of the multidrug-resistant isolates was susceptible to methicillin (cefepime). All *S. aureus* isolates were sensitive to vancomycin.

Table 1
Distribution of *S. aureus* among different sampling sites

Sampling category	Total number of samples (number & percentage of samples positive for <i>S. aureus</i>)	Sampling sites	Total number of samples (number & percentage of samples positive for <i>S. aureus</i>)
FTSs	64 (<i>n</i> = 12, 25.5%)	Bedside table handle	21 (8, 38.1%)
		IV pole grab area	14 (2, 14.3%)
		Room doorknob	7 (1, 14.3%)
		Room light switch	11 (1, 9.1%)
		Tray table	11 (0, 0%)
PISs	60 (<i>n</i> = 16, 34%)	Patient chairs	10 (1, 10%)
		Bed linen	20 (5, 25%)
		Bed rail	30 (10, 33.3%)
		Cardiac monitor	10 (1, 10%)
ME	59 (<i>n</i> = 12, 25.5%)	Oxygen supply unit	29 (9, 31%)
		Surgical aspirator	3 (1, 33.3%)
		Syringe pump machine	1 (1, 100%)
		Blood pressure apparatus	12 (0, 0%)
		Nebulizer	1 (0, 0%)
		Weight machine	1 (0, 0%)
		X-ray illuminator	2 (0, 0%)
		Ward sink	62 (7, 11.3%)
WSs	62 (<i>n</i> = 7, 11.3%)		

FTSs = Frequently touched surfaces; PISs = Patient's immediate surroundings; ME = Medical equipment; WS = Ward sinks

Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) to disinfectants

The MIC₉₀ ranges observed for CHX and CTAB were 2–256 μ g/ml, while for BAC MIC₉₀ range was 1–16 μ g/ml. The MBC₉₀ ranges observed for CHX and CTAB were 8–512 μ g/ml and 2–512 μ g/ml respectively, while for BAC MBC₉₀ range was 1 – 64 μ g/ml (Table 2). Majority of the of *S. aureus* isolates showed higher MIC₉₀ values that is, ≥ 64 μ g/ml for CHX (87.2%) as compared to BAC (0%) and CTAB (4.2%). For BAC, the isolates mainly displayed MIC₉₀ values of 1 μ g/ml (38.3%), 2 μ g/ml (23.4%) and 4 μ g/ml (19.1%). For CTAB, the isolates mainly displayed MIC₉₀ values of 1 μ g/ml (38.3%), 2 μ g/ml (23.4%) and 4 μ g/ml (19.1%).

Detection of *qac* genes in *S. aureus* isolates

Representative amplicons of each gene (*qacA*, *qacE* and *qac Δ E1*) detected on gel electrophoresis are shown in Figure 1A. The *qacA* gene was found in 11 isolates (23.4%). The *qacE* gene was detected in 14 (29.7%) isolates, while *qac Δ E1* was found in only 2 isolates (4.2%). MRSA isolates showed a higher frequency of *qac* genes (35%) as compared to Methicillin-sensitive *Staphylococcus aureus* (MSSA) (14.3%) isolates; however, it was statistically insignificant (*P* = .4) (Fig 1B). There was also no statistically significant association between the presence of *qac* genes and resistance to antibiotics tested against these isolates. Positive isolates with both *qacA* and *qacE* genes were most frequent (7%), followed by the *qacE* gene alone (4%). Isolates positive for *qac Δ E1* always carried the *qacE* gene. *S. aureus* isolates from the patient's immediate surroundings had the highest number of *qac* genes, followed by frequently touched surfaces.

Association between the presence of *qac* genes and MIC values

Isolates carrying the *qacA* and *qacE* genes had higher MIC₉₀ values to CHX, although it was not statistically significantly different from *qacA* and *qacE* negative *S. aureus* isolates. *qac Δ E1* positive *S. aureus* isolates had higher MIC₉₀ values to CHX, which were statistically significantly different from *qac Δ E1* negative *S. aureus* isolates (*P* = .02).

Isolates carrying *qacA* genes had statistically significantly higher MIC values to BAC (*P* = .01), similarly, isolates carrying *qacE* and *qac Δ E1* genes had higher MIC₉₀ values to BAC, however, the difference between *qacE/qac Δ E1* positive and *qacE/qac Δ E1* negative *S.*

Table 2
Comparison of MIC and MBC values to chlorhexidine (CHX), benzalkonium chloride (BAC), and cetrimonium bromide (CTAB) for *S. aureus* isolates with and without *qac* genes

Biocide	MIC90			Gene	MBC90			Number of isolates (n)			MIC90			MBC90		
	Median ($\mu\text{g/ml}$)	Range ($\mu\text{g/ml}$)	P-value		Median ($\mu\text{g/ml}$)	Range ($\mu\text{g/ml}$)	P-value				Median ($\mu\text{g/ml}$)	Range ($\mu\text{g/ml}$)	P-value			
CHX	128	2-256		<i>qacA</i>	256	8-512		11	Positive	128	32-128	256	128-256	.5		
				<i>qacE</i>				36	Negative	96	02-256	256	08-512			
				<i>qacE</i>				14	Positive	128	32-128	256	128-256	0.1		
				<i>qacE</i>				33	Negative	64	2-256	256	8-512			
				<i>qacE</i>				2	Positive	128	64-128	256	256-256	.02		
				<i>qacE</i>				45	Negative	96	2-256	256	8-512			
BAC	2	1-16		<i>qacA</i>	8	1-64		11	Positive	4	1-16	16	8-32	.01		
				<i>qacE</i>				36	Negative	2	1-16	8	1-64			
				<i>qacE</i>				14	Positive	4	1-16	12	4-32	.2		
				<i>qacE</i>				33	Negative	2	1-16	8	1-64			
				<i>qacE</i>				45	Positive	1	1-16	4	4-4	.1		
CTAB	4	2-256		<i>qacA</i>	16	2-512		11	Positive	8	2-256	32	8-512	.4		
				<i>qacE</i>				36	Negative	4	2-64	16	2-128			
				<i>qacE</i>				14	Positive	6	2-256	48	8-512	.2		
				<i>qacE</i>				33	Negative	4	2-64	16	2-128			
				<i>qacE</i>				2	Positive	10	4-16	48	32-64	.4		
				<i>qacE</i>				45	Negative	4	2-256	16	2-512			

A P-value of .05 was considered statistically significant.

aureus isolates were statistically insignificant. Isolates with *qacA*, *qacE*, and *qacE* had higher MIC values to CTAB however, the difference was statistically insignificant (Table 2).

For CHX, isolates with *qacA*, *qacE* and *qacE* genes showed comparable MBC values to *qacA*, *qacE*, and *qacE* negative isolates. For BAC, isolates carrying *qacA* genes had higher MBC values that were statistically significantly different from *qacA* negative isolates ($P = .003$). Similarly, MBC values were higher in isolates carrying *qacE* and *qacE* genes to BAC, however, the difference was statistically insignificant. All isolates with *qacA*, *qacE*, and *qacE* genes had higher MBC values to CTAB, however only *qacE* positive isolates had statistically significantly higher MBC values ($P = .03$) (Table 2).

DISCUSSION

In this study, 245 environmental samples were collected from a 1030-bed tertiary care hospital to determine the prevalence of *S. aureus* in the hospital environment. *S. aureus* was found to be present on 19.1% of the environmental surfaces. Other Pakistani studies revealed *S. aureus* contamination of hospital environmental surfaces at 28.7%,²¹ 32%,²² 29%,²³ and 40%.²⁴ However, in this study, methicillin resistance was found in a higher percentage (85.1%) of isolates than in another study from Pakistan (57.81%).²² This high prevalence of MRSA in the hospital setting might be due to natural selection, especially by natural transformation, which permits methicillin resistance to be transferred to methicillin-sensitive strains.²⁵ Furthermore, the greater frequency of *qac* genes in MRSA isolates in the current study compared to MSSA isolates may also have contributed to the selection pressure on MRSA isolates.

In this study, the highest number of *S. aureus* isolates were found in orthopedic (25%) and urology (20.8%) wards, of which 84% and 100% isolates were MRSA respectively. Several studies have reported *S. aureus* as one of the most frequently isolated pathogens in orthopedic wards and orthopedic surgical site infections.^{26,27,28} This high prevalence highlights the importance of having a basic understanding of the bacterium, its symptoms, and treatment options in the orthopedic community. A few studies have found urinary MRSA to be a potential HAI reservoir, resulting in occasional outbreaks of HA-MRSA in urology wards.^{29,30} This necessitates vigilant surveillance for successful outbreak prevention.

The sampling sites and frequency of exposure, particularly those that come into close contact with human hands, affect the types and distribution of bacteria isolated. For example, bedside table handles were found to be the most contaminated (38.1%) surfaces among the FTSS, possibly because of a high touch frequency from patients and attendants, who are the most uninformed in terms of infection and prevention and control. Similarly, among the PISs, the most contaminated surfaces were bed rails (33%), possibly due to the high frequency with which the staff who are responsible for shifting patients on beds and changing the bed linens come into contact with them. Most housekeeping workers are usually from low socioeconomic backgrounds, with limited education, awareness, and training in infection prevention and control, putting them and others at risk of acquiring HAIs.³¹ In short, bacterial contamination was more frequent on inanimate surfaces with direct hand contact by people who were maybe the most uninformed of IPC practices. The distribution of *S. aureus* isolates varied depending on the equipment used.¹¹ However, *S. aureus* was not isolated from blood pressure equipment, which is typically handled by nurses and doctors. Ward sinks, on the other hand, had lower levels of *S. aureus* contamination. This is consistent with previous studies, which found that *S. aureus* was more commonly isolated from PISs, MEs, and FTSS.¹¹ In this study, multi-drug-resistant (MDR) MRSA with resistance to combinations of oxacillin, fluoroquinolones, and macrolides was frequently isolated, suggesting the existence of selection pressure for this bacterium.

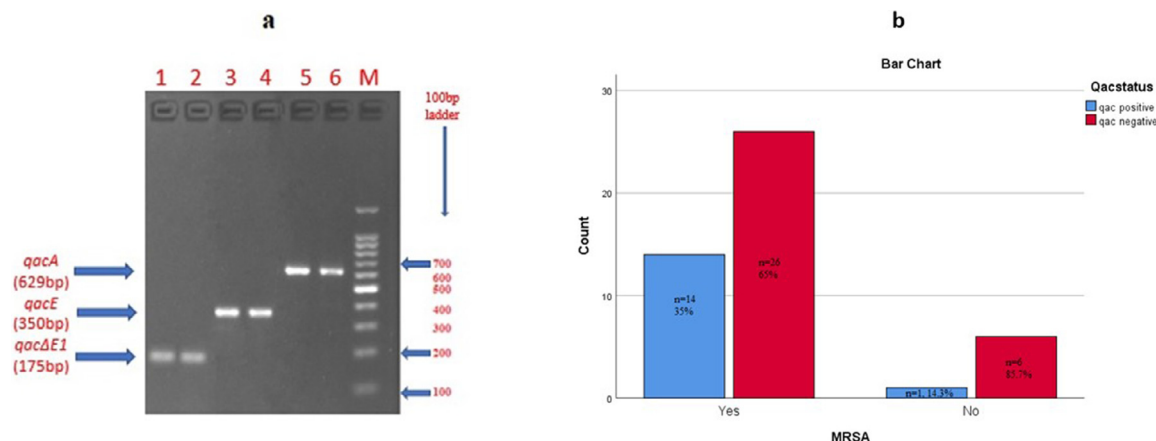


Fig 1. Detection of *qac* genes and their distribution in Methicillin-resistant and sensitive *S. aureus* isolates from hospital environment (A) Detection of *qacA*, *qacE*, and *qacΔE1* genes in *S. aureus* isolates by PCR agarose gel electrophoresis: 1-2 represent *qacΔE1*, 3-4 represent *qacE*, 5-6 represent *qacA* and M represents DNA marker size (100bp), (B) Distribution of *qac* genes in MRSA and MSSA isolates.

To our knowledge, this is the first study to report biocide susceptibility and the presence of the *qacA*, *qacE*, and *qacΔE1* genes in *S. aureus* isolated from a hospital environment in Pakistan. In 2013, there was only 1 study on *S. aureus* clinical isolates from 2 Peshawar tertiary care teaching hospitals that reported biocide susceptibility.³² The aforementioned study found lower MIC90 values to CHX (median = 0.25 μg/ml), however, the current study found substantially higher MIC90 values to CHX (median = 128 μg/ml). In the current study, the median BAC and CTAB MIC90 values were lower than in the above-mentioned study, but the MIC90 ranges were broader, with upper limits of 16 μg/ml and 256 μg/ml, respectively, compared to 3 μg/ml and 15 μg/ml.³² This shows that, even though the majority of the strains had lower MIC90 values, the current study found *S. aureus* strains with significantly higher MIC90 values in the hospital environment. Several studies have demonstrated decreased susceptibility to chlorhexidine,^{33,34,35} although the current study found the highest MIC90 values to date. This indicates that *S. aureus* isolates in the hospital environment may have been exposed to significant amounts of biocides, and strains with lower susceptibility are spreading into the hospital environment. However, the clinical significance of this finding is debatable because the concentration of biocides used in situ is typically much higher than the MICs, even for strains with reduced susceptibility. This is likely to become more important when other factors are taken into account, such as biofilm formation, which compromises the activity of the biocides. Besides these limitations, it is possible to determine proper disinfection concentrations for a facility's sanitization program using MIC determination and activity spectrum classification of disinfectants. Some other studies showed similar or lower MIC90 values to BAC and CTAB.^{2,36,37}

The prevalence of *qac* genes is frequently reported in *S. aureus* clinical isolates, however, very limited data is available on the presence of *qac* genes in environmental isolates with no studies from Pakistan. Disinfectants are routinely used in the hospital environment, and microorganisms come into regular contact with these substances. These genes confer reduced susceptibility to commonly used disinfectants, such as cetrimide, benzalkonium chloride and chlorhexidine. *qacA* gene is commonly associated with reduced susceptibility to disinfectants in staphylococci³⁵ and has been found in 0.9 to 83.3% of MRSA isolates from various countries,^{35,37} however, in the present study, the *qacE* gene was the most frequently isolated gene (29.7%) followed by *qacA* (23.4%), which is a rare finding. To our knowledge, only 1 study has described *qacE* genes in Gram-positive bacteria, including staphylococci,³⁸ although this observation has not been confirmed afterwards. However, the presence of *QacΔE1* in *S. aureus* has been confirmed in a few other studies.³⁹

Isolates carrying the *qacA*, *qacE*, and *qacΔE1* genes had higher MIC90 values, however, only the *qacA* and *qacΔE1* genes were statistically significantly associated with higher MIC90 values for BAC and CHX respectively. Only *qacA* and *qacΔE1* genes showed statistically significant differences in MBC90 values for BAC and CTAB respectively. Other studies have found a statistically significant association between the *qacA* gene and higher BAC and CHX MIC values.^{16,40,41,42} However, there is limited data to compare the presence of *qacE* and *qacΔE1* genes with MIC values in *S. aureus* isolates. Moreover, heterogeneity exists among several studies regarding the association of disinfectant resistance genes with reduced susceptibility to disinfectants. In our study, isolates with *qac* genes had higher MIC values than isolates without *qac* genes, but the difference was statistically insignificant. This statistically insignificant difference can be clinically significant⁴³ and could be due to the small sample size or other environmental factors.⁴⁴

In the current study, a few strains of *S. aureus* carrying *qacA* genes had lower MIC values to chlorhexidine than *qacA* negative strains. Measuring RNA expression of disinfectant resistance genes during exposure to disinfectants is one way to further investigate this variability. *Staphylococci* can have many disinfectant efflux mediated resistance genes, which are part of tightly regulated virulence systems. As a result, it is critical to understand how bacteria regulate and express different genes of these virulence systems in response to various environmental factors, as well as the broader implications of regulatory networks that could lead to efflux-based resistance.

CONCLUSION

The current study found significant MRSA contamination in the hospital environment. Their distribution in the hospital environment and the isolation of multiple MRSA isolates indicate that environment may be an important source of nosocomial pathogens. Furthermore, higher *qac* gene frequencies were found in MRSA isolates that also correlated with higher MIC/MBC values to different disinfectants in isolates carrying *qac* genes. The use of disinfectants is probably the best option for disease control in a post-antibiotic era, however, it is critical to fully understand the mechanisms of resistance to disinfectants to prevent their misuse to a point where the majority of nosocomial pathogens develop resistance to disinfectants. Selective pressure in the hospital environment allows such strains to persist and spread. Therefore, it is also important to keep track of the acquisition of resistance genetic elements. This study suggests that hospitals should implement policies and techniques to determine MICs of disinfectants against the most frequent types of environmental isolates until

scientists can fully understand these complicated disinfectant resistance processes.

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References

- Abreu AC, Tavares RR, Borges A, Mergulhão F, Simões M. Current and emergent strategies for disinfection of hospital environments. *J Antimicrob Chemother.* 2013;68:2718–2732.
- Vijayakumar R, Sandle T. A review on biocide reduced susceptibility due to plasmid-borne antiseptic-resistant genes—special notes on pharmaceutical environmental isolates. *J Appl Microbiol.* 2019;126:1011–1022.
- Kramer A, Schwebke I, Kampf G. How long do nosocomial pathogens persist on inanimate surfaces? A systematic review. *BMC Infect Dis.* 2006;6:130.
- Dancer SJ. How do we assess hospital cleaning? A proposal for microbiological standards for surface hygiene in hospitals. *J Hosp Infect.* 2004;56:10–15.
- Siddiqui AH, Koirala J. *Methicillin-resistant Staphylococcus aureus (MRSA)*. Treasure Island (FL), United States: StatPearls Publishing; 2018.
- Boyce JM. Modern technologies for improving cleaning and disinfection of environmental surfaces in hospitals. *Antimicrob Resist Infect Control.* 2016;5:1–10.
- Knapp L, Amézquita A, McClure P, Stewart S, Maillard JY. Development of a protocol for predicting bacterial resistance to microbicides. *Appl Environ Microbiol.* 2015;81:2652–2659.
- Guo W, Shan K, Xu B, Li J. Determining the resistance of carbapenem-resistant *Klebsiella pneumoniae* to common disinfectants and elucidating the underlying resistance mechanisms. *Pathog Glob health.* 2015;109:184–192.
- Cheatham S, Thapaliya D, Taha M, et al. Prevalence of *Staphylococcus aureus* and methicillin-resistant *S. aureus* on environmental surfaces in Ohio nursing homes. *Am J Infect Control.* 2019;47:1415–1419.
- Khan HA, Baig FK, Mehboob R. Nosocomial infections: Epidemiology, prevention, control and surveillance. *Asian Pac J Trop Biomed.* 2017;7:478–482.
- Phoon HY, Hussin H, Hussain BM, et al. Distribution, genetic diversity and antimicrobial resistance of clinically important bacteria from the environment of a tertiary hospital in Malaysia. *J Glob Antimicrob Resist.* 2018;14:132–140.
- Guh A, Carling P. *Options for Evaluating Environmental Cleaning*. National Center for Emerging and Zoonotic Infectious Diseases (U.S.). Division of Healthcare Quality Promotion. Centers for Disease Control and Prevention; 2010.
- Rose LJ, Noble-Wang J, Arduino MJ. Surface sampling. *Manual environ microbiol.* 2016;15:2–6.
- Cheesbrough M. *District Laboratory Practice in Tropical Countries*, part 2 Cambridge university press; 2005.
- CLSI. 31st ed. *Performance Standards for Antimicrobial Susceptibility Testing*. M100. Wayne, PA: Clinical and Laboratory Standards Institute; 2021.
- CLSI. *Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically*. 11th ed. Wayne, PA: Clinical and Laboratory Standards Institute; 2018.
- LaBreck PT, Bochi-Layec AC, Stanbro J, Dabbah-Krancher G, Simons MP, Merrell DS. Systematic analysis of efflux pump-mediated antiseptic resistance in *Staphylococcus aureus* suggests a need for greater antiseptic stewardship. *MSphere.* 2020;5:e00959–19.
- Barry AL, Craig WA, Nadler H, Reller B, Christine CS, Jana MS. National Committee for Clinical Laboratory Standards (NCCLS). M26-A: Methods for determining bactericidal activity of antimicrobial agents. *approved guideline*. 1999;19:8–19.
- Dashti AA, Jadaon MM, Abdulsamad AM, Dashti HM. Heat treatment of bacteria: a simple method of DNA extraction for molecular techniques. *Kuwait Med J.* 2009;41:117–122.
- Hassan MI, Al-Khafaji NS. Genotypic detection of disinfectant resistant genes among bacterial isolate of surgical site infections. *Ann Romanian Soc Cell Biol.* 2021:371–380. <https://www.annalsofscb.ro/index.php/journal/article/view/2472>.
- Shaheen A, Baqai R. Biofilm Formation by Environmental Microbes Isolated from Hospitals in Karachi, Pakistan. *ASRJETS.* 2016;15:240–251.
- Khan F, Ahmad B, Ahmed J, Yoo HS, Bashir S. Investigation of *Staphylococcus aureus*, prevailing in the environment of Khyber Teaching Hospital, Peshawar, Pakistan. *Pak J Pharm Sci.* 2018;31:429–437.
- Khattak SU, Bacha N, Lutfullah G, et al. Study of the genetic traits associated with antibiotic resistance in *Staphylococcus aureus* isolated from skin wards of Khyber Pakhtunkhwa, Pakistan. *Asian Pac J Trop Dis.* 2015;5:393–398.
- Zaib H, Kanwar R, Zafar N, Ali S. Prevalence and multidrug resistance profiles of several bacterial pathogens isolated from hospital inanimate surfaces in Faisalabad, Pakistan. *NRMJ.* 2019;3:526–534.
- Maree M, Thi Nguyen LT, Ohniwa RL, Higashide M, Msadek T, Morikawa K. Natural transformation allows transfer of SCCmec-mediated methicillin resistance in *Staphylococcus aureus* biofilms. *Nat commun.* 2022;13:1–4.
- Udobi CE, Obajuluwa AF, Onaolapo JA. Prevalence and antibiotic resistance pattern of methicillin-resistant *Staphylococcus aureus* from an orthopaedic hospital in Nigeria. *Biomed Res Int.* 2013;2013.
- Patel A, Calfee RP, Plante M, Fischer SA, Arcand N, Born C. Methicillin-resistant *Staphylococcus aureus* in orthopaedic surgery. *J Bone Joint Surg Br.* 2008;90:1401–1406.
- Kumari DN, Haji TC, Keer V, Hawkey PM, Duncanson V, Flower E. Ventilation grilles as a potential source of methicillin-resistant *Staphylococcus aureus* causing an outbreak in an orthopaedic ward at a district general hospital. *J Hosp Infect.* 1998;39:127–133.
- Takahashi S. MRSA in urology. In: Bjerklund Johansen TE, Wagenlehner FME, Matsumoto T, Cho YH, Krieger JN, Shoskes D, Naber KG, editors. *Urogenital Infections and Inflammations*. Berlin: German Medical Science GMS Publishing House; 2018.
- Coia JE, Duckworth GJ, Edwards DJ, et al. Joint working party of the british society of antimicrobial chemotherapy. guidelines for the control and prevention of methicillin-resistant *Staphylococcus aureus* (MRSA) in healthcare facilities. *J Hosp Infect.* 2006;63:S1–44.
- Vijaya R, Sharma A, Daga A, Gopichandran L. Effectiveness of structured training programme on infection control measures among housekeeping workers in selected intensive care units at tertiary hospital. *JoNSP.* 2020;10:51–56.
- Ahmad B, Urbas F, Jamil J, Ahmed J, Bashir S. Biocides susceptibility pattern and phenotypic detection of Efflux pump in *Staphylococcus aureus* isolates from two tertiary hospitals of Pakistan. *Afr J Microbiol Res.* 2013;7:3171–3178.
- Horner C, Mawer D, Wilcox M. Reduced susceptibility to chlorhexidine in staphylococci: is it increasing and does it matter? *J Antimicrob Chemother.* 2012;67:2547–2559.
- Kampf G. Acquired resistance to chlorhexidine—is it time to establish an ‘antiseptic stewardship’ initiative? *J Hosp Infect.* 2016;94:213–227.
- Htun HL, Hon PY, Holden MT, Ang B, Chow A. Chlorhexidine and octenidine use, carriage of qac genes, and reduced antiseptic susceptibility in methicillin-resistant *Staphylococcus aureus* isolates from a health care network. *Clin. Microbiol. Infect.* 2019;25:1154–1161.
- Kampf G. Benzalkonium chloride. *Antiseptic Stewardship. Biocide Resistance and Clinical Implications*. 1. Switzerland AG: Springer Nature; 2018:259–370.
- Zaki ME, Bastawy S, Montasser K. Molecular study of resistance of *Staphylococcus aureus* to antiseptic quaternary ammonium compounds. *J. Glob. Antimicrob. Resist.* 2019;17:94–97.
- Kazama H, Hamashima H, Sasatsu M, Arai T. Distribution of the antiseptic-resistance gene qacE delta 1 in gram-positive bacteria. *FEMS Microbiol Lett.* 1998;165:295–299.
- Ignak S, Nakipoglu Y, Gurler B. Frequency of antiseptic resistance genes in clinical staphylococci and enterococci isolates in Turkey. *Antimicrob Resist Infect Control.* 2017;6:88.
- Conceição T, Coelho C, de Lencastre H, Aires-de-Sousa M. High prevalence of biocide resistance determinants in *Staphylococcus aureus* isolates from three African countries. *Antimicrob Agents Chemother.* 2016;60:678–681.
- Vali L, Dashti AA, Mathew F, Udo EE. Characterization of heterogeneous MRSA and MSSA with reduced susceptibility to chlorhexidine in Kuwaiti hospitals. *Front Microbiol.* 2017;8:1359.
- Bes TM, Perdigão-Neto L, Martins RR, et al. *Susceptibility to Chlorhexidine and Mupirocin Among Methicillin-Resistant Staphylococcus Aureus Clinical Isolates from A Teaching Hospital*. 63. Revista do Instituto de Medicina Tropical de São Paulo; 2021:e27.
- Ranganathan P, Pramesh CS, Buyse M. Common pitfalls in statistical analysis: Clinical vs statistical significance. *Perspect Clin Res.* 2015;6:169.
- Cremieux A. Factors affecting the bactericidal action of disinfectants. Implications for selection of resistant strains. *Drugs Exp Clin Res.* 1986;12:899–903.